

DECODE function - is used to calculate and return expressions according to certain equality conditions

Syntax is:

```
DECODE (VARIABLE, condition1, expression1, condition2, expression2, ....., condition_n,
expression_n, value n+1 default)
```

The CASE expression - is used to calculate and return expressions depending on certain various conditions in which comparison operators can be used, not only the equality operator.

Syntax is:

```
CASE [VARIABLE]
WHEN condition1 THEN expression1
WHEN condition2 THEN expression2
.....
WHEN condition_n THEN expression_n
ELSE default value n+1
END
```

Or if comparisons are used on different variables or columns, the following syntax can be used:

```
CASE
WHEN VARIABLE condition1 THEN expression1
WHEN VARIABLE condition2 THEN expression2
.....
WHEN VARIABLE condition_n THEN expression_n
ELSE default value n+1
END
```

Exercises

1. Calculate the agents' commission according to the job they occupy:

- 0.1% of the value of the orders if the job is SA_REP
- 0.2% of the value of the orders if the job is SA_MAN
- 0.3% of the value of the orders if the job is AD_PRES

For the other jobs the commission will be 0.

```
SELECT first_name, job_id,
case upper(job_id)
WHEN 'SA_REP' THEN 0.1
WHEN 'SA_MAN' THEN 0.2
WHEN 'AD_PRES' THEN 0.3
ELSE 0
END commission
FROM
employees;
```

```
SELECT first_name, job_id,
case
WHEN upper(job_id)='SA_REP' THEN 0.1
WHEN upper(job_id)='SA_MAN' THEN 0.2
WHEN upper(job_id)='AD_PRES' THEN 0.3
ELSE 0
END commission
FROM employees;
```

If this commission is applied to the value of the orders (SUM (quantity * unit_price)) the query becomes:

```
SELECT first_name, job_id,
(CASE UPPER(job_id)
WHEN 'SA_REP' THEN 0.1
WHEN 'SA_MAN' THEN 0.2
WHEN 'AD_PRES' THEN 0.3
ELSE 0
END) * sum(quantity*unit_price)
commission_values
FROM employees e, orders o, order_items oi
WHERE e.employee_id=o.sales_rep_id AND
o.order_id=oi.order_id
GROUP BY first_name, job_id;
```

```
SELECT first_name, job_id,
(CASE
WHEN upper(job_id)='SA_REP' THEN 0.1
WHEN upper(job_id)='SA_MAN' THEN 0.2
WHEN upper(job_id)='AD_PRES' THEN 0.3
ELSE 0
END) * sum(quantity*unit_price)
commission_values
FROM employees e, orders o, order_items oi
WHERE e.employee_id=o.sales_rep_id AND
o.order_id=oi.order_id
GROUP BY first_name, job_id;
```

	FIRST_NAME	JOB_ID	COMMISSION_VALUES
1	Danielle	SA_REP	12824.95
2	Nanette	SA_REP	17197.31
3	Christopher	SA_REP	11421.57
4	Allan	SA_REP	15629.62
5	Sarath	SA_REP	66173.45
6	Janette	SA_REP	20261.76
7	Louise	SA_REP	8823.84
8	Oliver	SA_REP	13441.52
9	Lindsey	SA_REP	15116.72

2. Calculate the discount (DC) for customers differently as follows:

- if the customer has completed 1 order then DC = 10%;
- if he concluded 2 orders then DC = 15%;
- if you have completed more than 3 orders then DC = 20%.

```
SELECT cust_first_name, count(o.order_id) no_of_orders,
CASE WHEN COUNT(o.order_id) = 1 THEN 0.1
WHEN COUNT(o.order_id) = 2 THEN 0.15
WHEN COUNT(o.order_id)>=3 THEN 0.2
ELSE 0 END discount
FROM customers c, orders o
WHERE c.customer_id=o.customer_id
GROUP BY cust_first_name;
```